

Homework for week 1

Exercises from the Kreyszig textbook:

- 11.1 problems 14 and 19,
- 11.2 problem 17

Compute the Fourier sine and cosine coefficients of the 2π -periodic general rectangular pulse with width $2w$ ($w < \pi$) and height h :

$$f(x) = \begin{cases} h & -w < x < w, \\ 0 & \text{otherwise.} \end{cases}$$

$$f(x) = f(x + 2\pi)$$

Bonus: Consider the limit of a very narrow and tall pulse. Take the limit of $w \rightarrow 0$ and $h \rightarrow \infty$ while keeping the area under the pulse $A = 2wh$ constant. Express the Fourier coefficients with A .